

AI-Driven Protein Design with Diffusion and Language Models

Abstract

AI-driven protein design has moved from a largely expert-curated, physics-heavy optimization problem to a data-driven generative modeling discipline that increasingly integrates protein language models, geometric diffusion models, flow-matching methods, inverse-folding networks, and structure-prediction oracles into unified design stacks. The central technical shift is that modern systems do not merely score candidate sequences or predict structures; they learn distributions over sequences, backbones, side-chain environments, functions, and multimodal annotations, and then sample or steer those distributions under user-specified objectives. Protein language models exploit the scale of sequence corpora to learn evolutionary regularities and local fitness structure, whereas diffusion and flow-based models provide especially strong control over three-dimensional geometry, motif scaffolding, symmetry, interfaces, and all-atom contexts. The most successful contemporary pipelines are therefore increasingly hybrid: they combine sequence priors from language models, controllable structure generation from diffusion or flow models, inverse folding to translate structures into realizable sequences, and structure/function predictors for ranking or iterative refinement [8–12, 19–34, 35–52, 53–63, 64–125, 149–156, 163–200]. ¹

This survey synthesizes the field through July 2026 with an emphasis on the methodological interface between diffusion models and protein language models. The main conclusions are threefold. First, diffusion and flow methods now dominate controllable backbone generation because they naturally accommodate equivariance, conditional sampling, and manifold-aware geometry. Second, language models remain the most flexible machinery for sequence priors, zero-shot fitness estimation, retrieval, function conditioning, and multimodal text/annotation integration. Third, the frontier is no longer a competition between the two families, but a convergence toward multimodal, all-atom, closed-loop systems that couple generation with inverse folding, retrieval, experimental feedback, and biosecurity-aware governance. At the same time, serious obstacles remain: benchmark leakage, the mismatch between foldability and function, poor handling of dynamics and disorder, expensive wet-lab validation, and dual-use risks in increasingly capable open protein generators [44–52, 65–66, 75, 89–125, 118–148, 157–162]. ²

Introduction

Protein design has long pursued the inverse of structure prediction: rather than inferring form from sequence, it seeks sequences and structures that realize specified biochemical functions. Classical achievements from Rosetta and directed evolution established that de novo folds, interfaces, and ligand binders are possible, but those workflows often required carefully engineered objective functions, handcrafted constraints, and substantial experimental iteration [1–7]. The arrival of high-accuracy learned structure predictors, especially AlphaFold, RoseTTAFold, ESMFold, and later AlphaFold 3 and RoseTTAFold All-Atom, transformed this landscape by making structural inference far more accessible and by providing oracles that can be embedded directly into generative design loops [8–14]. ³

In parallel, protein language modeling established that large self-supervised models trained on raw sequences recover nontrivial biological regularities without explicit structural supervision. Early transfer-learning benchmarks such as TAPE, sequence representation models such as UniRep and Bepler-Berger, large masked models such as ESM and ProtTrans, autoregressive families such as ProGen and ProtGPT2, and multimodal successors such as ESM3 showed that evolutionary statistics encoded in natural sequences are sufficient to support mutation-effect prediction, inverse folding, sequence generation, and, increasingly, direct design [15–34, 163–180]. This body of work reframed proteins as a native machine-learning modality rather than a special-case downstream application of bioinformatics tools.

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Diffusion-based protein generation introduced an orthogonal capability. Instead of learning only the discrete statistics of amino-acid strings, protein diffusion models learn to reverse corruption processes over 3D coordinates, residue frames, or joint sequence–structure representations. Chroma, RFDiffusion, FrameDiff, Genie-family models, ProteinGenerator, and subsequent sparse, all-atom, and flow-matching variants demonstrated that geometric generation can be conditioned on motifs, symmetries, interfaces, ligands, text, and functional hints, often with impressive wet-lab validation. This matters because many design objectives are natively geometric: active-site placement, interface complementarity, oligomeric symmetry, loop closure, pocket formation, and scaffoldability are not easily expressed in sequence space alone [53–63, 64–88, 118–136].

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The present survey argues that the most important contemporary question is not whether diffusion or language models are “better,” but how they factorize the protein design problem. Language models are strongest at capturing sequence priors, family semantics, mutational fitness, and flexible conditioning from text or metadata; diffusion and flow models are strongest at high-dimensional geometric control and structural diversity; inverse-folding models bridge generated geometry back to sequence; and predictor-oracles operationalize designability, affinity, or activity in closed-loop optimization. Accordingly, this survey focuses on their complementarity, their integration into end-to-end design pipelines, the emerging evaluation standards required to compare them fairly, and the challenges that still prevent reliable function-first protein engineering at scale [35–52, 64–125, 137–162].

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Methods

A useful unifying view is that protein design systems differ primarily in their **representation**, **generative objective**, and **conditioning operator**. Representations may be sequence-only, residue-frame based, backbone-atom based, all-atom, surface-aware, or multimodal with text/function tokens. Generative objectives may be autoregressive next-token prediction, masked reconstruction, discrete diffusion, continuous diffusion on Euclidean or Riemannian spaces, or flow matching. Conditioning may be architectural, prompt-like, retrieval-based, classifier-guided, reward-guided, or post hoc through external oracles. Modern systems increasingly mix these axes rather than committing to a single design primitive [8–12, 19–34, 53–63, 64–88, 97–111, 149–156].

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Protein language models. Sequence-only language models remain foundational because protein data are overwhelmingly more abundant in sequence than in experimentally solved structure. Masked models such as ESM, MSA Transformer, ProteinBERT, and ProtTrans learn bidirectional context and serve as strong feature extractors; autoregressive models such as ProGen, ProGen2, ProtGPT2, and later large multitask or long-context variants directly define generative priors over entire amino-acid sequences [15–27, 163–180]. These models are valuable not simply because they generate plausible strings, but because they encode mutational tolerance, evolutionary syntax, and family-level semantics that can be repurposed for fitness estimation, property prediction, and conditional sampling [28–34, 115–117, 167–180].

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A key conceptual distinction inside the language-model family is between **unconditional pretraining** and **design-time steering**. Early generative use emphasized unconditional or lightly conditioned sequence generation, but more recent work increasingly biases generation with prompts, classifiers, retrieval, sequence-family fine-tuning, or experimental preference signals. ZymCTRL, conditional enzyme-language models, inference-time retrieval in Tranception-like systems, function-guided steering, and multimodal text-to-protein frameworks such as ProteinDT and related natural-language design systems exemplify this shift [27, 32–34, 98–117, 173–182, 188–197]. In other words, the field is moving from “sample likely proteins” to “sample proteins likely to satisfy an external objective.” ⁹

Inverse folding as the engineering bridge. The practical bridge between structure generation and realizable proteins is inverse folding. Graph-based protein design models, GVP-based encoders, ProteinMPNN, PiFold, LM-assisted inverse folding, CarbonDesign, LigandMPNN, and multimodal all-atom variants convert a target backbone or complex into sequences predicted to fold to that geometry [35–52, 183–200]. This family is essential because many successful design stacks first generate candidate structures with diffusion or flow models, then apply inverse folding, then validate the resulting sequence–structure pair with AlphaFold-class predictors or self-consistency checks. LigandMPNN and function-preserving multimodal redesign methods further show that conditioning on ligands, metals, nucleic acids, or side-chain environments is critical when the design target depends on nonprotein context [44–47, 183–189]. ¹⁰

A deeper methodological point is that inverse folding changes what “success” means. Natural sequence recovery is convenient, but it is not identical to designability or function. Preference-optimization approaches, multimodal retrieval systems, and newer all-atom inverse-folding objectives therefore seek better alignment between recovery metrics and downstream design success. This is one reason the inverse-folding literature is converging with the alignment literature from large language models: both are asking how to reshape a pretrained generator toward task-relevant outputs rather than merely optimize likelihood on existing data [48–52, 186–197]. ¹¹

Backbone diffusion and geometric generation. Structure-space diffusion models established the modern generative regime for protein backbones. FrameDiff introduced frame-based denoising over residue geometry; RFDiffusion adapted RoseTTAFold-like inductive biases to iterative backbone denoising and became a workhorse for motif scaffolding, binder design, and symmetric assemblies; Chroma formulated protein generation as programmable Bayesian inference under external constraints; and later systems broadened scale, speed, and controllability [64–80]. The success of these methods rests on a fortunate fit between the mathematics of diffusion and the geometry of proteins: denoising can be defined over coordinates or frames, equivariance preserves rigid-motion symmetries, and conditioning can be injected at every step of the reverse process. ¹²

Flow matching and related continuous-time generative methods now play a closely related role. In theory, flow-based methods offer an alternative parameterization of generative transport that can reduce sampling cost and improve numerical stability; in practice, FoldFlow, FrameFlow, Proteína and later partially latent, few-step, or physics-guided variants reflect the field’s search for greater throughput without sacrificing designability [61–63, 69–80]. This acceleration matters because protein design is rarely a one-shot generation task. Real workflows often require thousands to millions of candidate trajectories, followed by inverse folding, structural ranking, developability filtering, and sometimes docking or experimental triage. Faster geometric generators therefore directly affect the scalability of scientific discovery campaigns [71–80, 129–136]. ¹³

Sequence diffusion and discrete denoising. Because protein sequences live on a discrete alphabet, a parallel line of work adapts diffusion-like ideas to categorical or latent sequence spaces. EvoDiff, DPLM and DPLM-2, diffusion on language-model encodings, TaxDiff, AMP-Diffusion, and discrete inverse-folding

variants show that denoising-based sequence generation can be competitive with or complementary to autoregressive decoding [89–109, 190–195]. The appeal is twofold. First, denoising can support non-left-to-right generation and therefore global refinement. Second, diffusion can more naturally accommodate multimodal conditioning when the output must satisfy multiple simultaneous constraints, such as foldability, text semantics, taxonomy, antimicrobial activity, or a desired response curve as in ForceGen [89–96, 104, 190]. ¹⁴

The distinction between **structure-space diffusion** and **sequence-space diffusion** is now central to protein design methodology. Structure-space diffusion is generally superior for motif placement, topology control, and interface shape. Sequence-space diffusion is often more natural for soft compositional objectives, family-conditioned generation, or multistate sequences where the same chain must support multiple folds or contexts. ProteinGenerator made this contrast explicit by positioning sequence-space diffusion as a complement to backbone-space RFDiffusion rather than a replacement for it [65, 75–84, 89–109]. This complementarity is a recurring theme across the literature: geometry controls what proteins can look like; sequence diffusion and language models help specify which realizations are biologically coherent or functionally enriched. ¹⁵

Multimodal and hybrid systems. The most important recent trend is hybridization. ESM3-style multimodal masked generation, DPLM-2, trimodal protein search models, text-guided ProteinDT, natural-language protein design, CLIP-style protein editing, structure language models, and agentic design systems all attempt to unify sequence, structure, function, and text in a single representational space [32, 91, 97–107, 173, 180–182]. Hybridization also appears architecturally: some systems use language-model embeddings as the latent substrate of diffusion; others use diffusion-generated structures as prompts to inverse-folding language models; still others use external predictors and retrievers to guide sampling without retraining the base generator [92, 98–111, 188–197]. The net effect is a gradual collapse of the old boundaries between “protein LM,” “inverse folder,” “diffusion backbone generator,” and “design oracle.” ¹⁶

Guidance and alignment. Once a base generator exists, the central research question becomes how to steer it. The literature now includes parameter-updating methods such as supervised family fine-tuning, preference optimization, and reward-guided adaptation, as well as parameter-fixed methods such as plug-and-play classifier guidance, retrieval augmentation, importance reweighting, Feynman–Kac or particle-based steering, and experimental feedback loops [48–50, 110–117, 188–197]. This alignment perspective is especially fruitful because it recognizes that natural proteins are not the optimal target distribution for many engineering tasks. Desired solutions may be evolutionarily rare, underrepresented in public databases, or even absent from natural sequence space altogether. A well-designed steering layer is therefore not an optional add-on; it is often the difference between sampling biologically plausible proteins and sampling proteins that satisfy an actual engineering brief. ¹⁷

From models to pipelines. The field’s strongest demonstrations increasingly come from integrated design stacks rather than isolated architectures. RFDiffusion and RFDiffusion2 backbone generators paired with ProteinMPNN or LigandMPNN support motif scaffolding, binder design, and active-site construction; sequence-space diffusion enables multistate design; ligand-aware inverse folding enables enzyme and sensor design; and multimodal language systems promise direct conditioning from text descriptions or high-level functional instructions [37, 44, 65, 75, 98–125]. Application papers now report increasingly ambitious tasks, including enzyme active-site scaffolding, binders to intrinsically disordered targets, sequence-specific DNA-binding design, neosurface targeting, and small-molecule-binding proteins created from scratch [118–125, 120–123]. That breadth is strong evidence that AI protein design is becoming a platform technology rather than a niche methodology. ¹⁸

Challenges and Prospects

The first major challenge is **evaluation**. Benchmarking in protein design is unusually fragile because leakage can occur at the sequence, structure, family, scaffold, ligand-scaffold, or design-template level. Random train/test splits can therefore massively overestimate generalization. Newer benchmarks such as ProteinInvBench, ProteinBench, PFMBench, FLIP, ProteinGym-style mutational datasets, PINDER, PLINDER, PoseBusters, and function-guided benchmark suites have improved the situation, but the field still lacks universal consensus on task definitions, the proper balance between likelihood-like and function-like metrics, and how to compare sequence and structure generators under shared constraints [51–52, 137–148]. Most importantly, self-consistency under a structure predictor is necessary but not sufficient: a protein can appear foldable *in silico* and still fail experimentally due to expression, aggregation, dynamics, context dependence, or incorrect active-site chemistry. ¹⁹

The second challenge is **the function gap**. Foldability, novelty, and geometric plausibility are significantly easier objectives than catalytic proficiency, conformational switching, allostery, disordered-region recognition, or tight small-molecule binding. Recent work on multistate diffusion, designed binders to intrinsically disordered proteins, atom-level active-site scaffolding, and conformation-aware language models is beginning to address this gap, but most current generators still learn static snapshots more easily than free-energy landscapes or kinetic mechanisms [84, 101, 107, 118–125]. The broader lesson is that proteins are not merely static 3D objects; they are thermodynamic and often multistate systems. Truly reliable design will need stronger integration of dynamics, uncertainty, and environment-specific behavior into both training objectives and evaluation protocols [2, 11–12, 79, 152]. ²⁰

The third challenge is **data quality, scaling, and bias**. Sequence resources are abundant but unevenly distributed across clades and protein families. Structural data are smaller, biased toward experimentally tractable proteins, and historically sparse for complexes, disorder, liganded states, and transient conformers. Synthetic pseudo-labeling with AlphaFold-class predictors and large-scale structure databases mitigates some scarcity, but it also risks amplifying the biases of the underlying predictors. Reviews of protein language models and generative protein design increasingly emphasize that scaling alone does not solve these issues, and the broader machine-learning literature warns that indiscriminate recursive training on model-generated data can degrade the statistical quality of future models [14, 19–21, 32–34, 149–156, 162]. Compute is a related bottleneck: all-atom multimodal systems, repeated self-consistency checks, and complex-aware sampling remain expensive, slowing iteration in practical design campaigns [75, 97, 154–156, 173–182]. ²¹

The fourth challenge is **responsible deployment**. The same models that can accelerate therapeutic enzyme or binder design could also lower barriers to harmful protein engineering. Current scholarly discussion therefore extends beyond open-source release questions to include access control, watermarking, calibrated guidance, red-teaming benchmarks, and policy frameworks for dual-use governance [157–161]. This is not peripheral to method design: capability, usability, and steerability all affect risk. A model that is easy to condition with natural language, capable of optimizing multiple biochemical objectives, and paired with powerful inverse-folding and ranking oracles is scientifically exciting precisely because it is operationally potent. Consequently, biosecurity cannot remain a downstream compliance layer; it must shape evaluation, model release, and API design from the outset [127–128, 157–161]. ²²

The clearest prospect is the emergence of **closed-loop multimodal design systems**. The likely mature architecture for the field is a stack in which a multimodal foundation model proposes candidate sequence/structure/function plans; a diffusion or flow generator instantiates geometric realizations; an inverse-folder designs compatible sequences in all-atom context; retrieval and experimental datasets

steer optimization toward rare objectives; and structure, docking, or activity predictors provide calibrated feedback. Such systems will probably operate interactively, support text and symbolic prompts, and natively reason over sequence, structure, ligands, nucleic acids, and annotations. The deepest open research question is whether these models can move from generating plausible proteins to generating reliably functional molecular mechanisms. Progress will likely depend less on scaling any single architecture and more on improving the interfaces among language priors, geometric generators, oracles, data curation, and experimental feedback [11–12, 32, 44, 65, 71–80, 91–125, 149–156, 198–200].

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Conclusion

The contemporary literature strongly suggests that diffusion models and protein language models occupy complementary, not competing, positions in protein design. Diffusion and flow-based methods currently provide the most powerful tools for programmable geometric generation, especially when topology, active-site placement, symmetry, or interface shape are central. Protein language models provide the strongest sequence priors, the most flexible interfaces for text or annotation conditioning, and increasingly capable zero-shot or retrieval-augmented estimates of fitness and function. Inverse folding and structure-prediction oracles then connect these modalities into operational pipelines. The frontier of AI-driven protein design, therefore, is the design of systems of models rather than the scaling of any single model family [32, 37, 44, 65–66, 75, 91–125, 149–156].

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The field has already crossed an important threshold: it can create nontrivial proteins that are experimentally testable under geometrically and functionally meaningful constraints. Yet the decisive milestone is still ahead. Reliable protein engineering will require evaluation protocols that resist leakage, models that understand dynamics and context rather than only static folds, and governance mechanisms proportionate to the growing power of generative biomolecular systems. If these issues are addressed with the same seriousness that has driven recent architectural innovation, the union of diffusion and language modeling may become the core computational substrate for next-generation enzyme engineering, therapeutic design, biomaterials, and molecular discovery [118–148, 157–162].

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